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Properties of Materials
Lecture 12

Dr. Alkis Tsamis
at441@leicester.ac.uk

CTGU/UoL JEP, September 2024

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Lecture 8: Bonding Between Atoms, Learning Objectives, Bonding Energies, Primary Bonding, Ionic Bonding, Covalent Bonding, Metallic Bonding, Crystal Structures, Body-Centred Cubic, Face-Centred Cubic, Secondary Bonding, Van der Waals Forces, Hydrogen Bonding, Bonding Energies and Properties, Bonding Types and Properties, Concepts and Terminology (4)
Lecture 9: Recap from Last Time, Task
Exercise Seminar 2 / Lecture 10: Packing of Atoms in Solids (1), Learning Objectives, The Structure of Metals and Ceramics (1), Unit Cell Parameters, Face-Centred Cubic (fcc), Hexagonal Close-Packed (hcp), Crystal Structures - Metals, Crystal Structures - Ceramics, Concepts and Terminology (5)
Lecture 11: Packing of Atoms in Solids (2), Learning Objectives, Atomic Packing, Atomic Packing - bcc Structures, Atomic Packing Factor - bcc, Atomic Packing - fcc Structures, Atomic Packing Factor - fcc, Atomic Packing - hcp Structures, Atomic Packing Factor - hcp, Tips for Calculating APF for hcp, Concepts and Terminology (6)
Lecture 12: The Physical Basis of Young's Modulus, Learning Objectives, Moduli of Crystals, Rubbers and the Glass Transition Temperature, Composite, Composite Terminology, Fibre-Reinforced Composites, Rule of Mixtures (1)

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The Physical Basis of Young's Modulus

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Recap from last time...

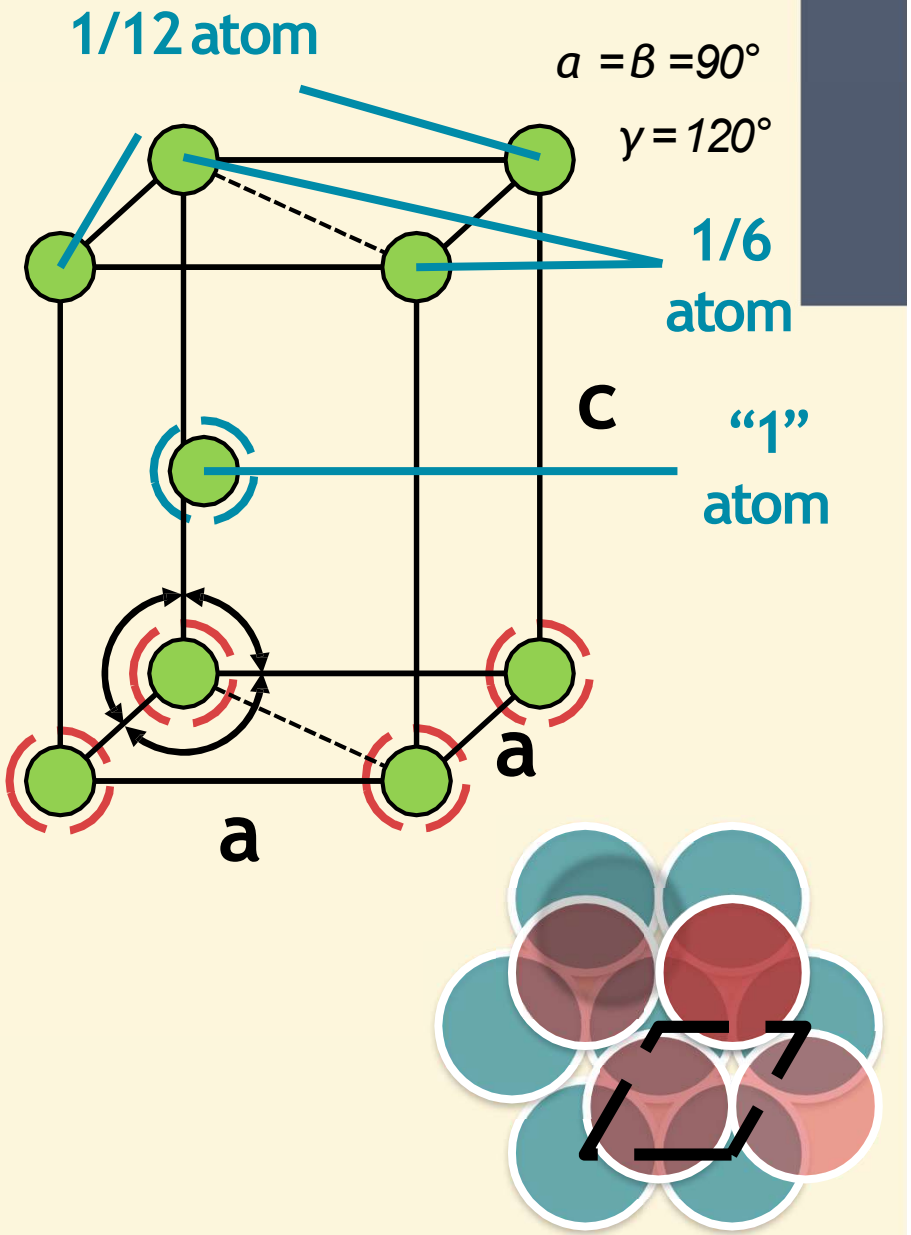
Atomic packing factor – hcp (1)

$$APF = \frac{N_{atoms} V_{atoms}}{V_{unit\ cell}}$$

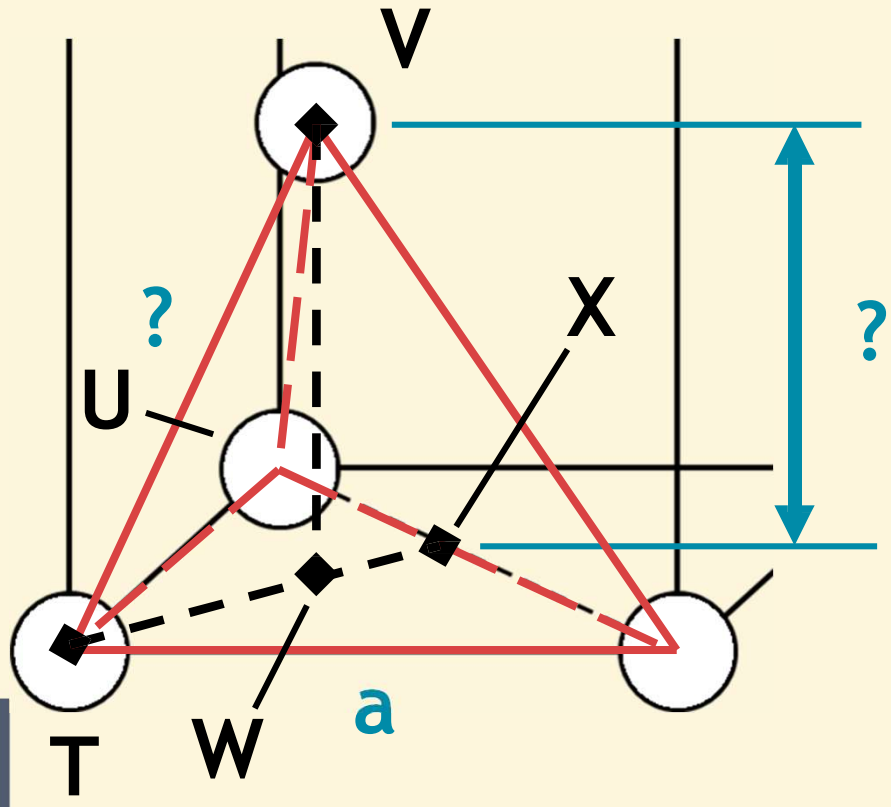
$$N_{atoms} = 4(1/12) + 4(1/6) + 1(1) = 2$$

$$V_{atoms} = \frac{4}{3}\pi R^3$$

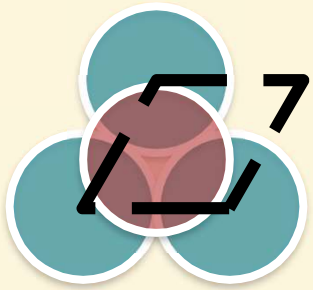
$$V_{unit\ cell} = A_{base}(c) = a^2 \cos(30°) c$$



Atomic packing factor – hcp (2)



$a = \beta = 90°$
$\gamma = 120°$



$$APF = \frac{N_{atoms} V_{atoms}}{V_{unit\ cell}}$$

$$N_{atoms} = 2$$

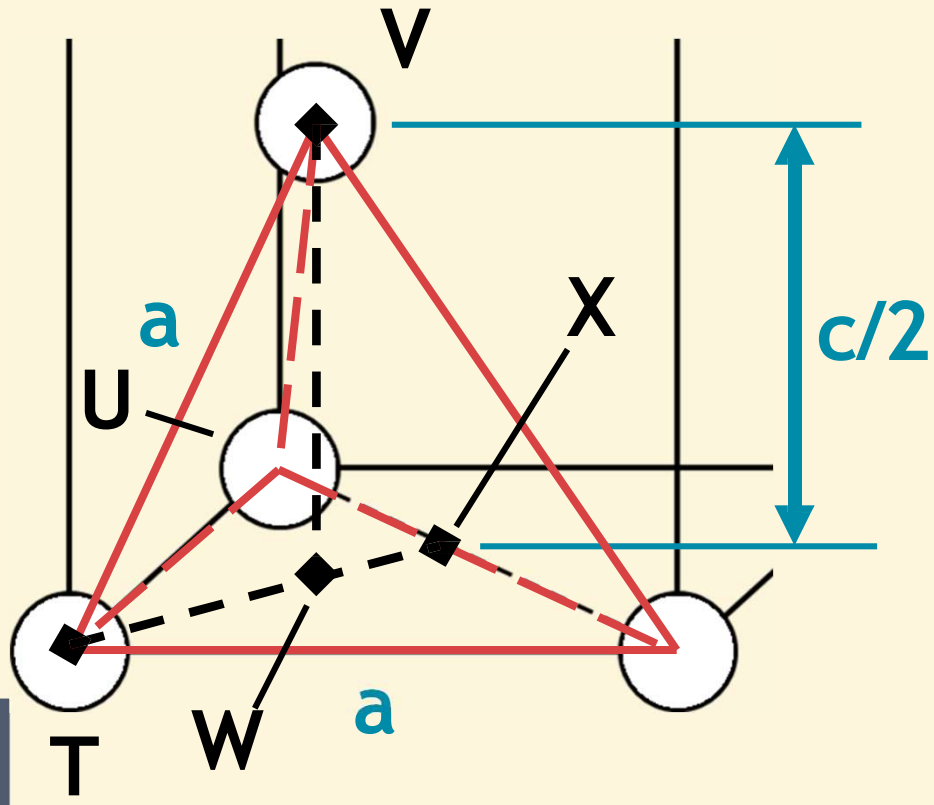
$$V_{atoms} = \frac{4}{3}\pi R^3$$

$$\overline{WV} = ?$$

Distance between **T** and **V**?

$$\overline{TV} = ?$$

Atomic packing factor – hcp (3)



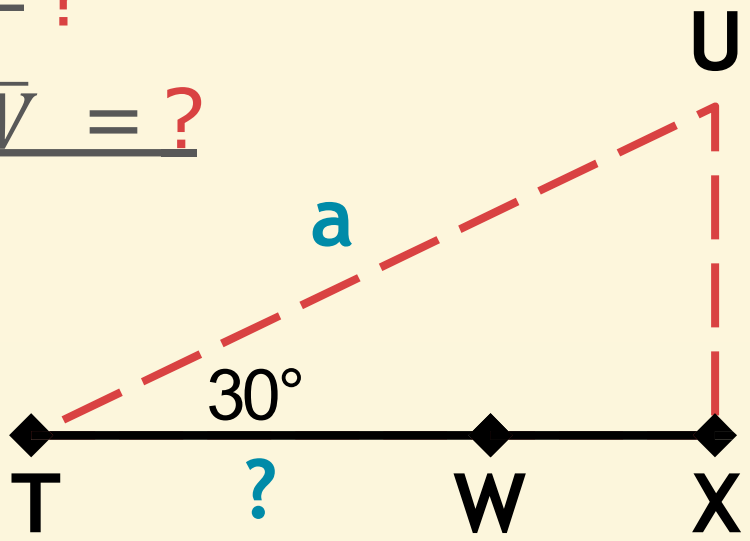
$\overline{WV} = \frac{c}{2}$; $\overline{TV} = a = 2R$

Find **TW**:

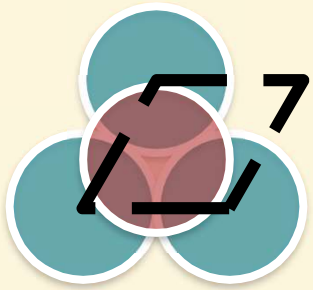
$\overline{TW} = ?$

$\overline{TX} = ?$

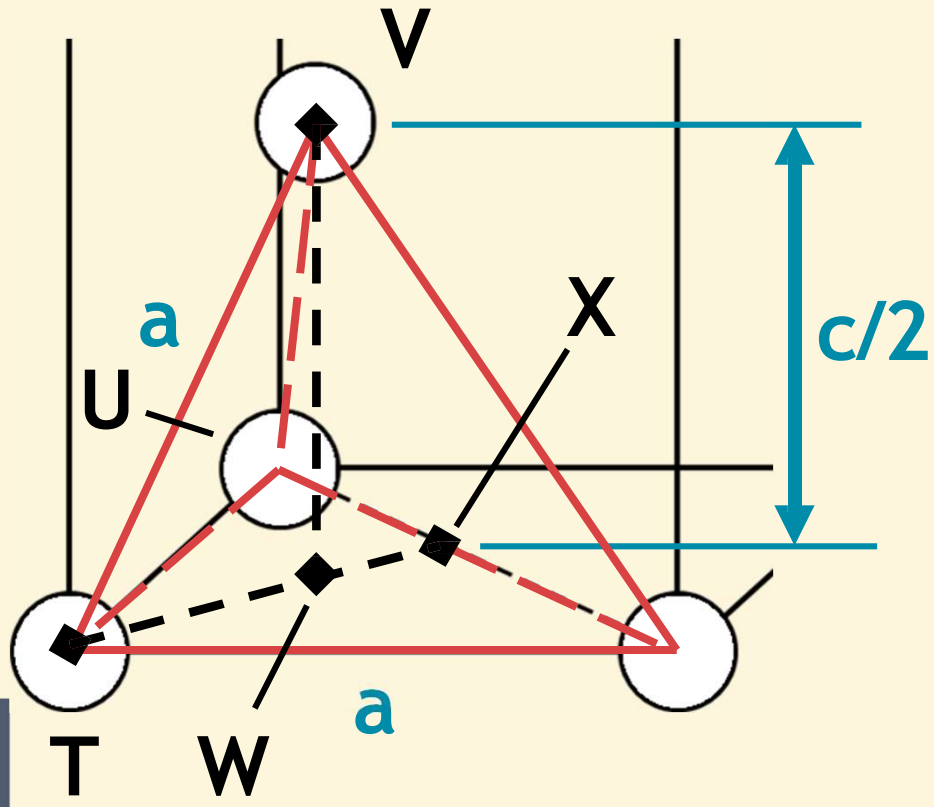
$\therefore \overline{TW} = ?$



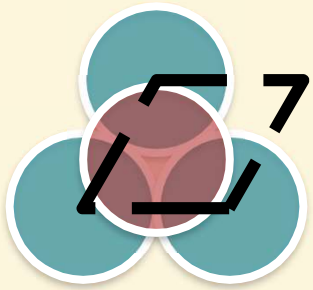
$a = \beta = 90°$
$\gamma = 120°$



Atomic packing factor – hcp (4)



$a = \beta = 90°$
$\gamma = 120°$



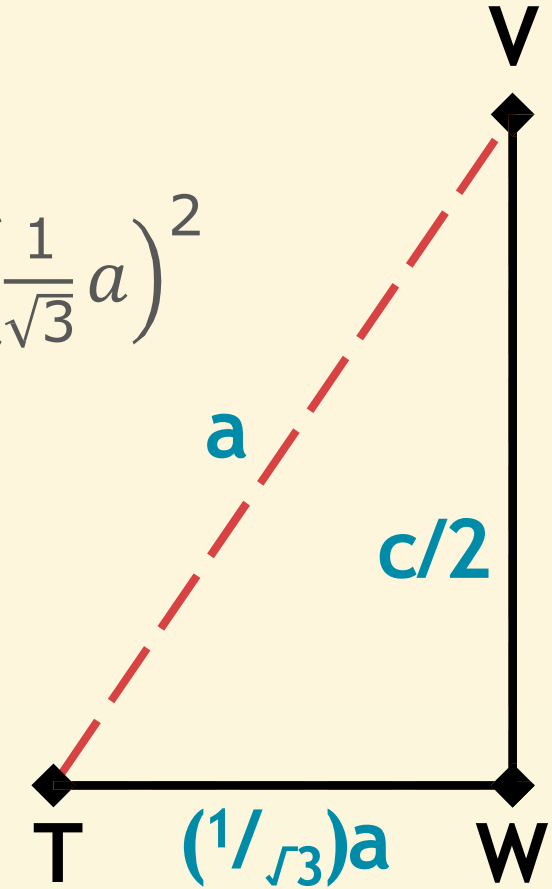
$$\overline{WV} = \frac{c}{2} \; ; \; \overline{TV} = a = 2R$$

$$\overline{TW} = \frac{1}{\sqrt{3}}a$$

$$\left(\frac{c}{2}\right)^2 = a^2 - \left(\frac{1}{\sqrt{3}}a\right)^2$$

$$\Rightarrow c = \sqrt{\frac{8}{3}}a$$

$$\therefore c = 4\sqrt{\frac{2}{3}}R$$



Atomic packing factor – hcp (5)

$$N_{atoms} = 2$$

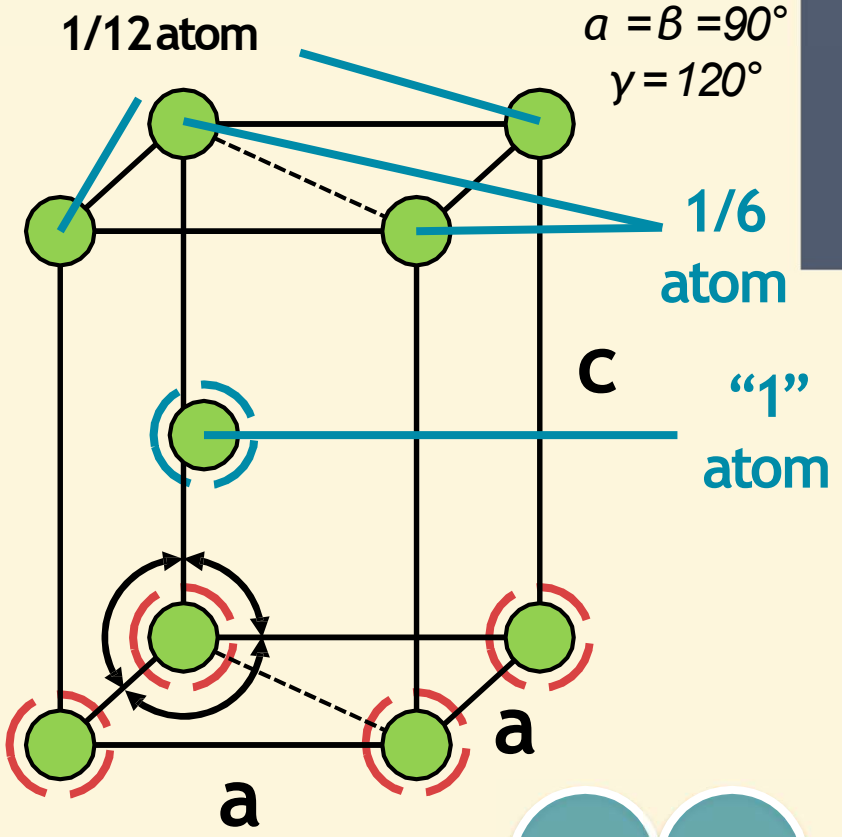
$$V_{atoms} = \frac{4}{3}\pi R^3$$

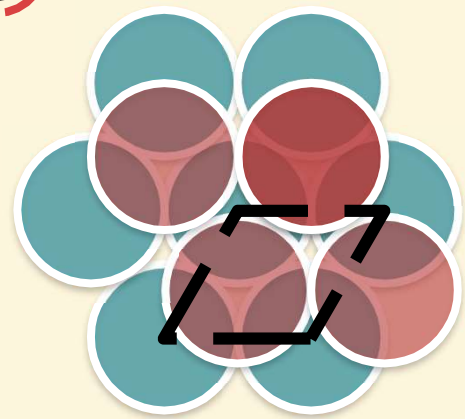
$$V_{unit\ cell} = a^2 c \cos 30°$$

$$a = 2R;\ c = 4\sqrt{\frac{2}{3}}R$$

$$\Rightarrow V_{uc} = (?)(?)(?)$$

$$\therefore V_{uc} = ???R^3$$





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Atomic packing factor – hcp (6)

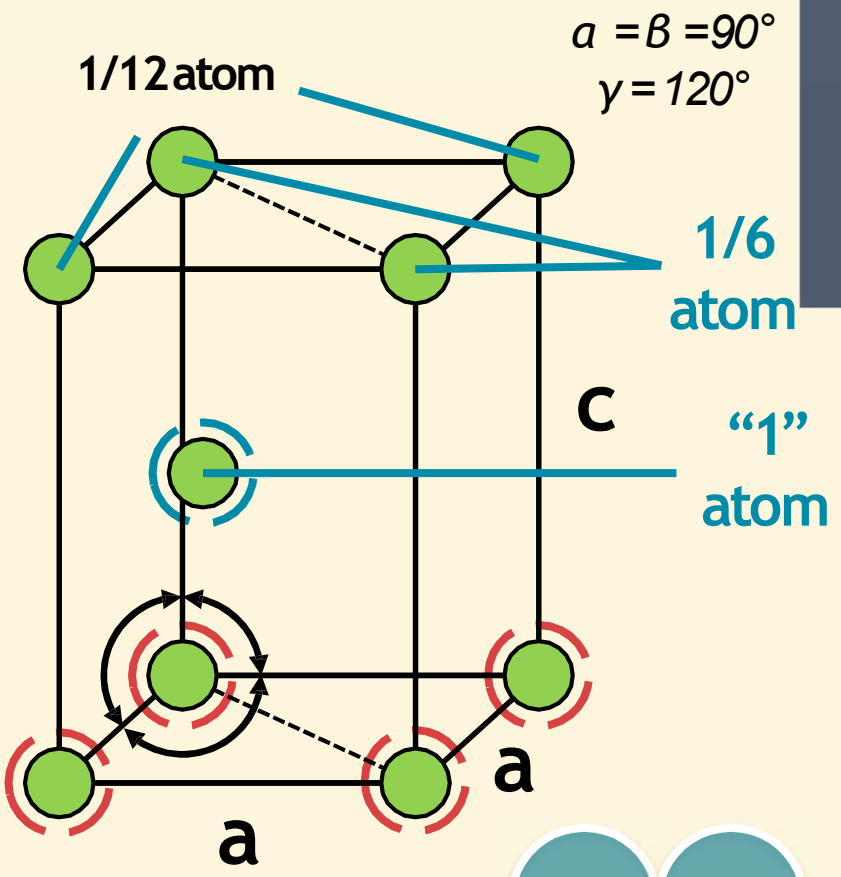
$$N_{atoms} = 2$$

$$V_{atoms} = \frac{4}{3}\pi R^3$$

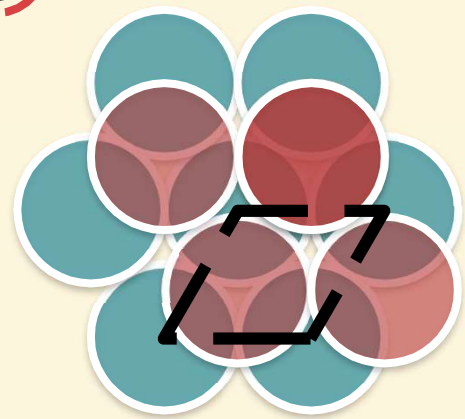
$$V_{uc} = 8\sqrt{2}R^3$$

$$APF = \frac{N_{atoms}V_{atoms}}{V_{unit\ cell}} = \frac{?\left(\frac{4}{3}\pi R^3\right)}{?}$$

$\therefore$ $APF = ???$







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Learning objectives

- To bring together the factors underlying the moduli of materials.

Moduli of crystals

- Atoms in **crystals** are held together by bonds which behave like little springs. Stiffness of one of these bonds as

$$S_0 = \left(\frac{d^2U}{dr^2}\right)_{r=r_0}$$

- For small strains, spring constant of the bond stays 0.
- The **force between a pair of atoms**, stretched apart to a distance r - r_0 is

$$F = S_0(r - r_0)$$

Moduli of crystals

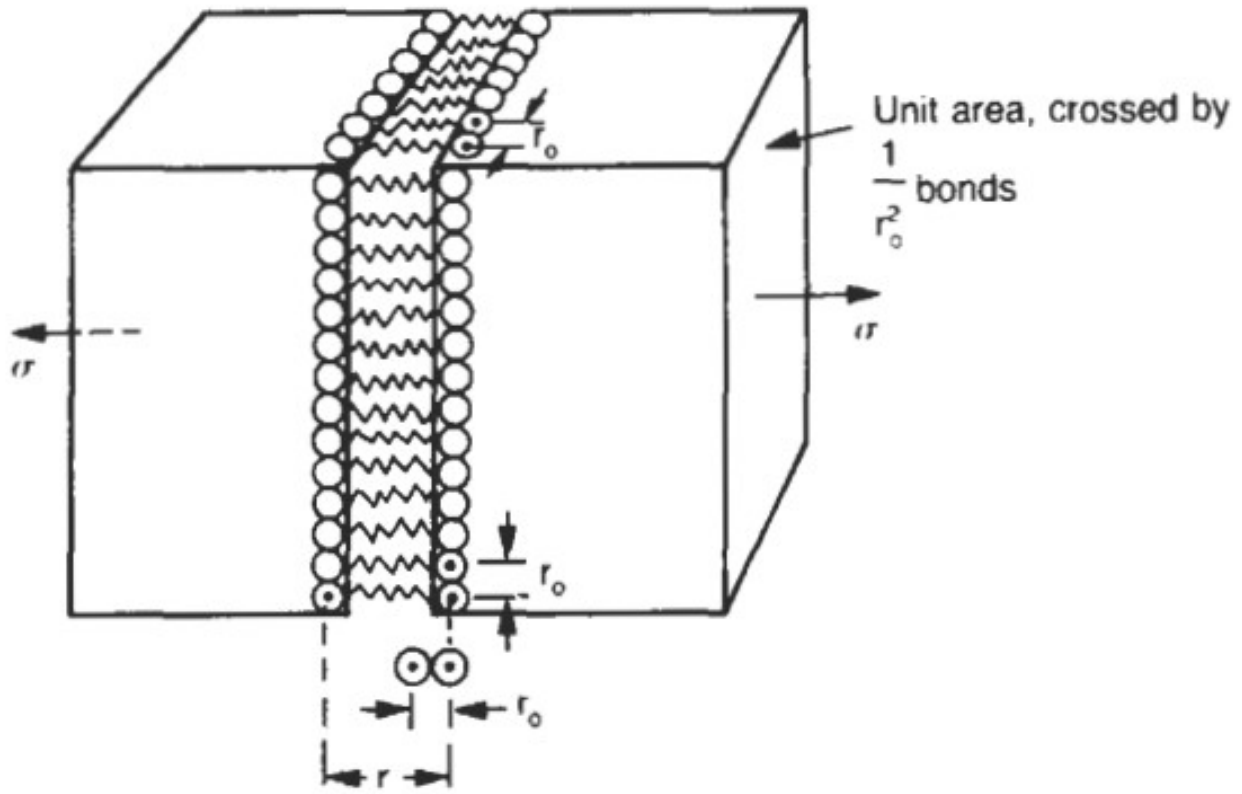


Fig. 6.1. The method of calculating Young's modulus from the stiffnesses of individual bonds.

Moduli of crystals

- Now, the **total force** exerted across unit area, if the two planes are pulled apart a distance ($r - r_0$) is defined as the stress σ, with

$$\sigma = NS_0(r - r_0)$$

- N is the number of bonds/unit area, equal to $1/r^2{}_0$, (since $r^2{}_0$ is the average area-per atom). **Converting displacement** $(r - r_0)$ into strain ϵ_n, by dividing by the initial spacing, r_0, so that

$$\sigma = \frac{S_0}{r_0}\epsilon_n$$

Moduli of crystals

- **Young's modulus**, then, is just

$$E = \frac{\sigma}{\epsilon_n} = \frac{S_0}{r_0}$$

Table 6.1

Bond type	S_o *(N m^{-1})*	*E (GPa); from eqn. (6.5) (with $r_0 = 2.5 \times 10^{-10}$ m)*
Covalent, e.g. C–C	50–180	200–1000
Metallic, e.g. Cu–Cu	15–75	60–300
Ionic, e.g. Na–Cl	8–24	32–96
H-bond, e.g. H_2O–H_2O	2–3	8–12
Van der Waals, e.g. Polymers	0.5–1	2–4

Rubbers and the glass transition temperature

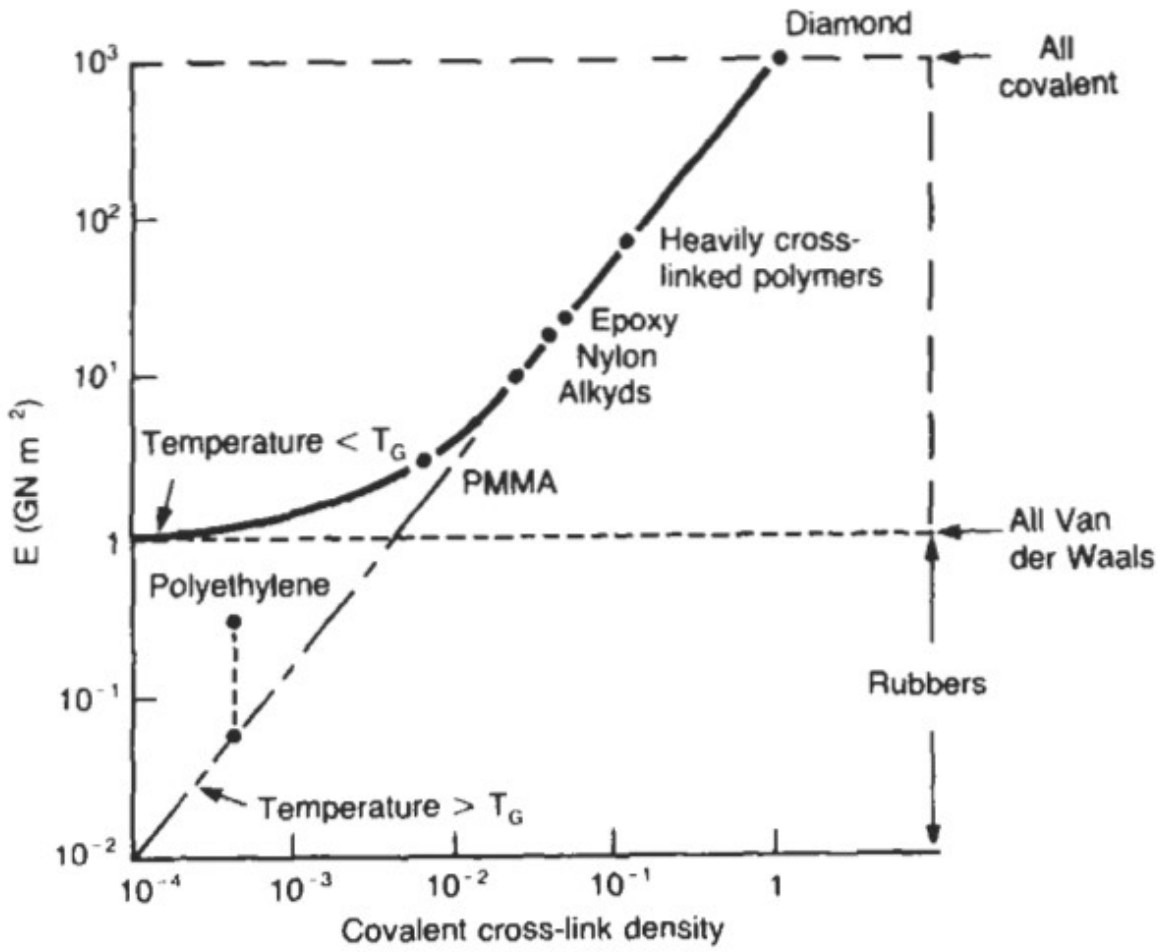


Fig. 6.2. How Young's modulus increases with increasing density of covalent cross-links in polymers, including rubbers above the glass temperature. Below T_G, the modulus of rubbers increases markedly because the Van der Waals bonds take hold. Above T_G they melt, and the modulus drops.

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Composite

- Composite materials are **multi-material**… err… materials!
  - Composites may be better described as '**structures**' rather than materials in their own right
  - Formed of different constituent parts that are 'fastened' together

- Benefit from the **combined material properties** of their constituents

- **Fibre-reinforced composites** are a common example of the benefits of composite structures:
  - e.g. GFRP (Glass Fibre Reinforced Polymers), CFRP (Carbon Fibre), KFRP (Kevlar Fibre)

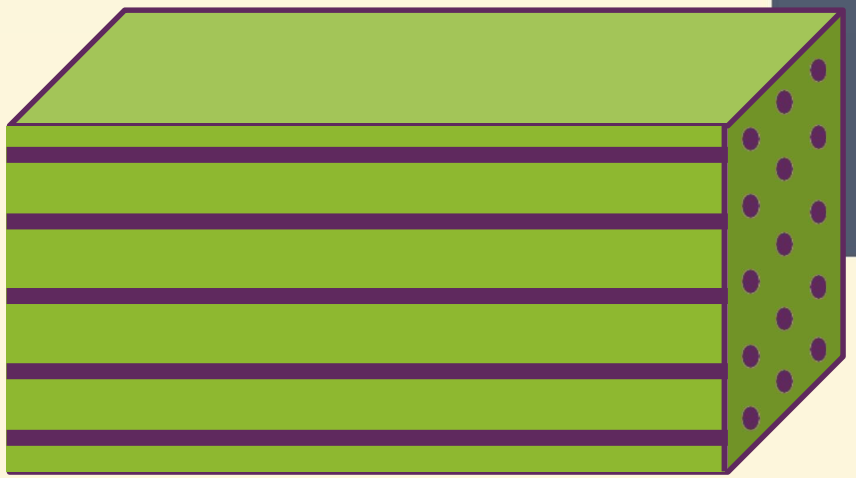
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Composite terminology

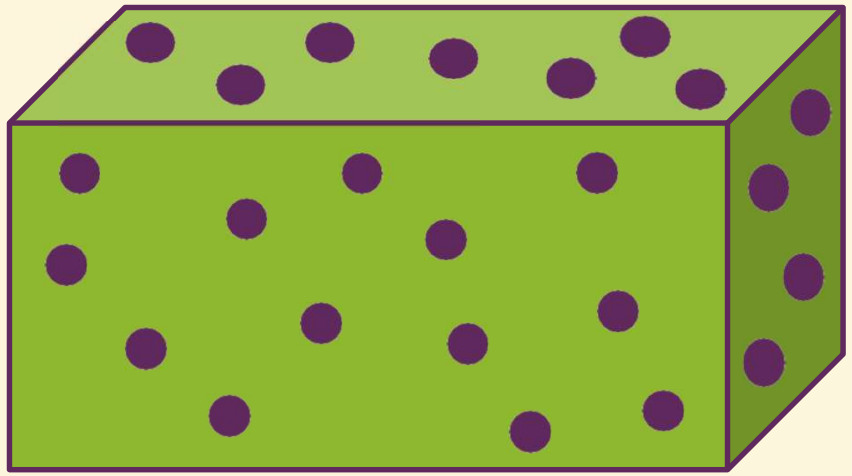
- **Matrix**:
  - 'Bulk' component of the composite structure

- **Fibre-reinforced** composites:
  - 'Supporting' component – long strands, usually specifically oriented

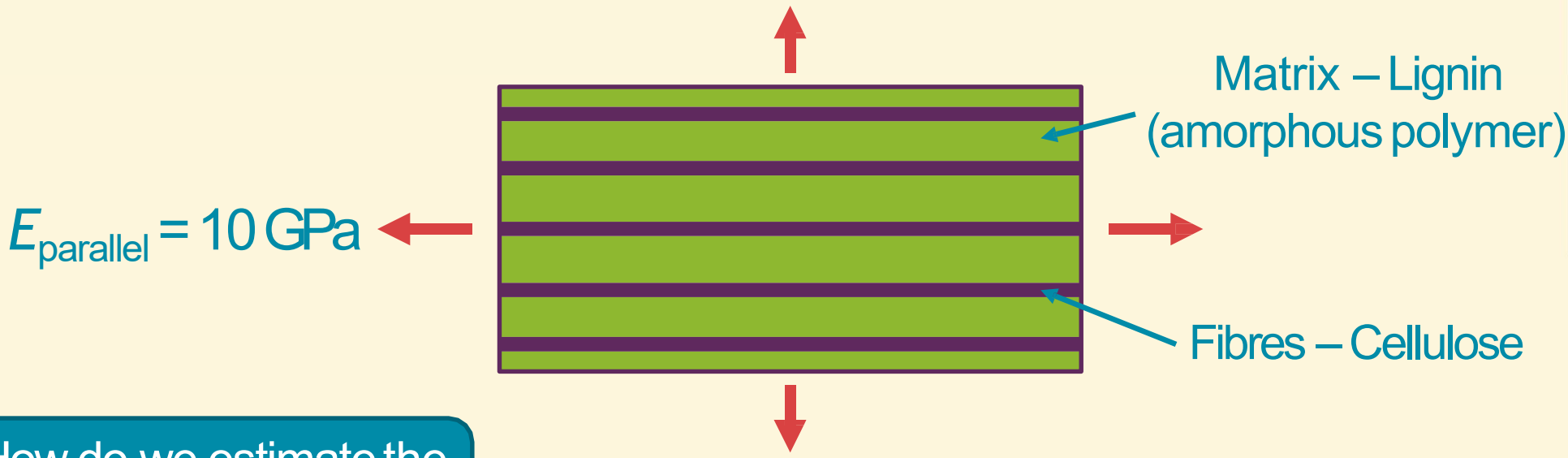
- **Particulate** composites (covered elsewhere):
  - 'Supporting' component – small particles, randomly distributed



Fibre-reinforced composite



Particulate composite



How do we estimate the strength of composites?

Fibre-reinforced composites

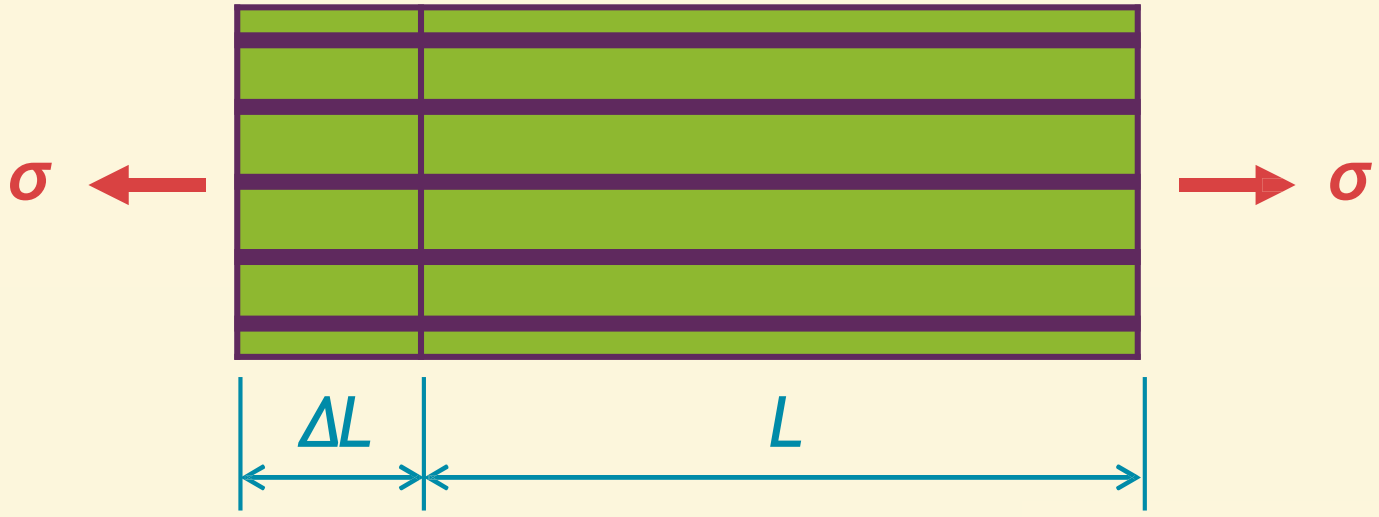
- Benefits include modulus much greater than that of their matrices and typically **very high specific stiffness** (E/ρ)
- Commonly very anisotropic – e.g. wood

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Rule of Mixtures

Rule of mixtures (1)

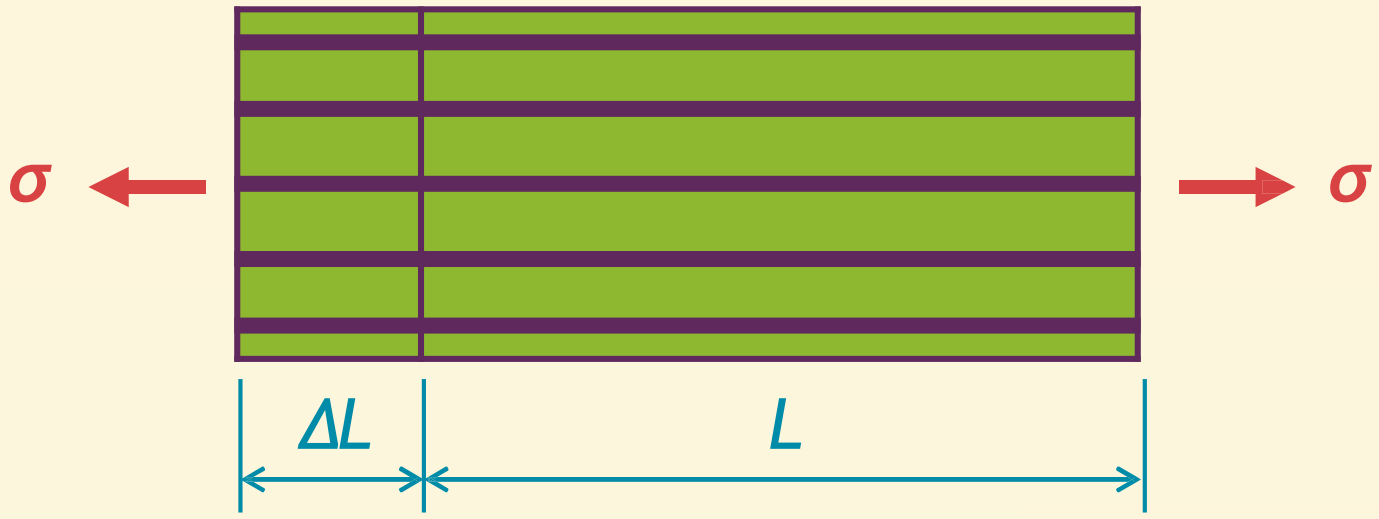
- Estimating the modulus of composites **(upper limit)**:
  - Subject composite to a known stress, **parallel** to the fibre direction
  - As fibres and matrix are bonded, they stretch by the same amount –e.g they have the **same strain, ε_n**
  - Assuming **equal proportions** of fibres and matrix, $\sigma = \left(\sigma_f + \sigma_m \right)/2$



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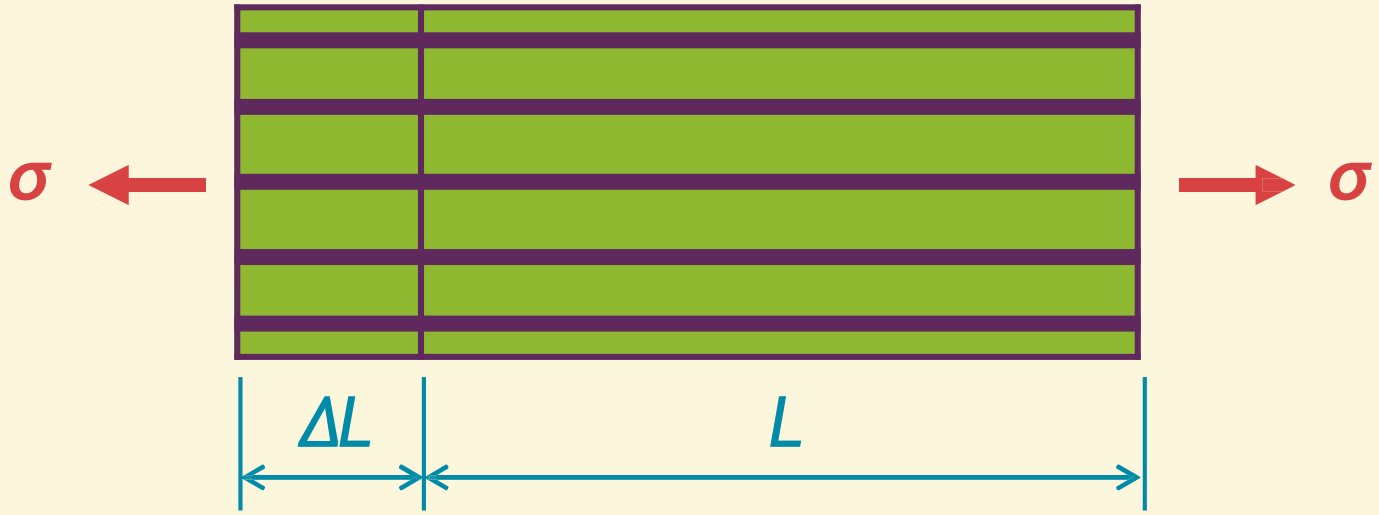
Rule of mixtures (2)

- In equal proportions, $\sigma = \left(\sigma_f + \sigma_m \right)/2$
- In un-equal proportions, $\sigma = V_f \sigma_f + \left(1 - V_f \right) \sigma_m$
  - V_f = volume fraction of fibres
- Since $\sigma = E \varepsilon_n$:
  - $\sigma = \left[E_f V_f + E_m \left(1 - V_f \right) \right] \varepsilon_n$



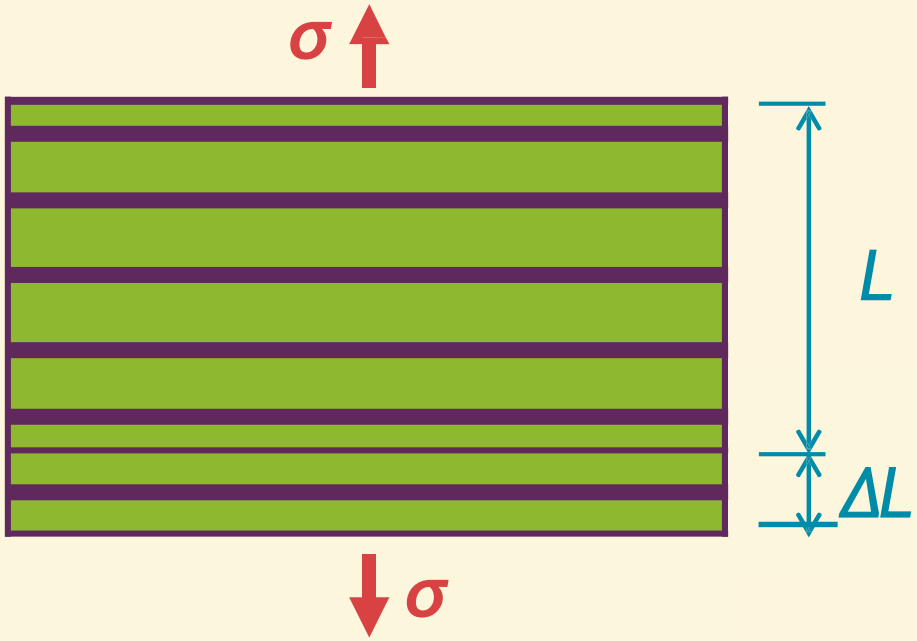
Rule of mixtures (3)

- $\sigma = \left[E_f V_f + E_m \left(1 - V_f\right)\right] \varepsilon$
- As $E_{composite} = \frac{\sigma}{\varepsilon}$
  - $E_{composite} = E_f V_f + E_m \left(1 - V_f\right)$
- **Upper estimate** for the composite modulus:
  - Need to consider loading perpendicular to fibres



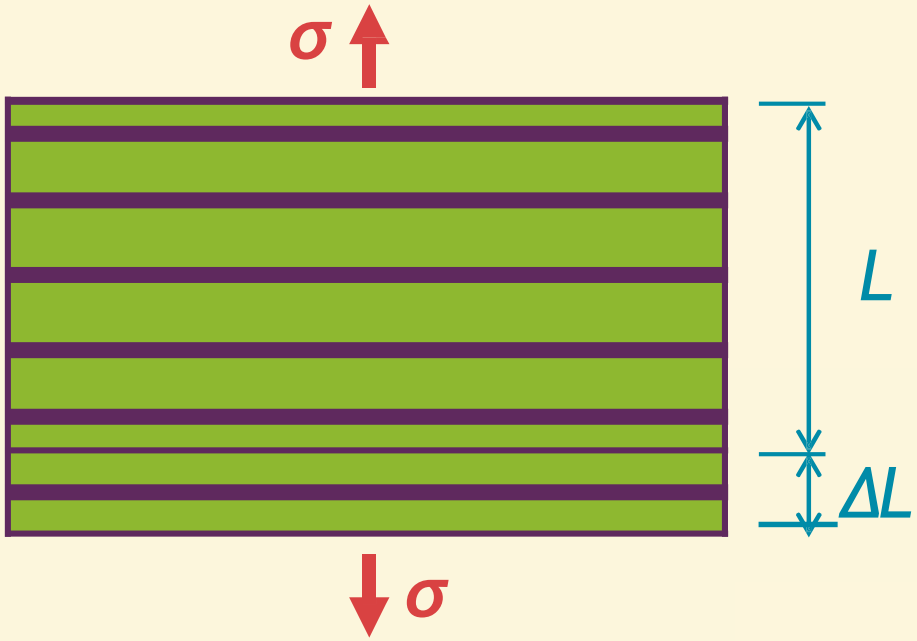
Rule of mixtures (4)

- Estimating the modulus of composites **(lower limit)**:
  - Subject composite to a known stress, **perpendicular** to the fibre direction
  - In this direction, it is reasonable to assume that they share the **same stress** and therefore have a weighted nominal strain
    - In **equal proportions**, $\varepsilon_n = (\varepsilon_{n,f} + \varepsilon_{n,m})/2$



Rule of mixtures (5)

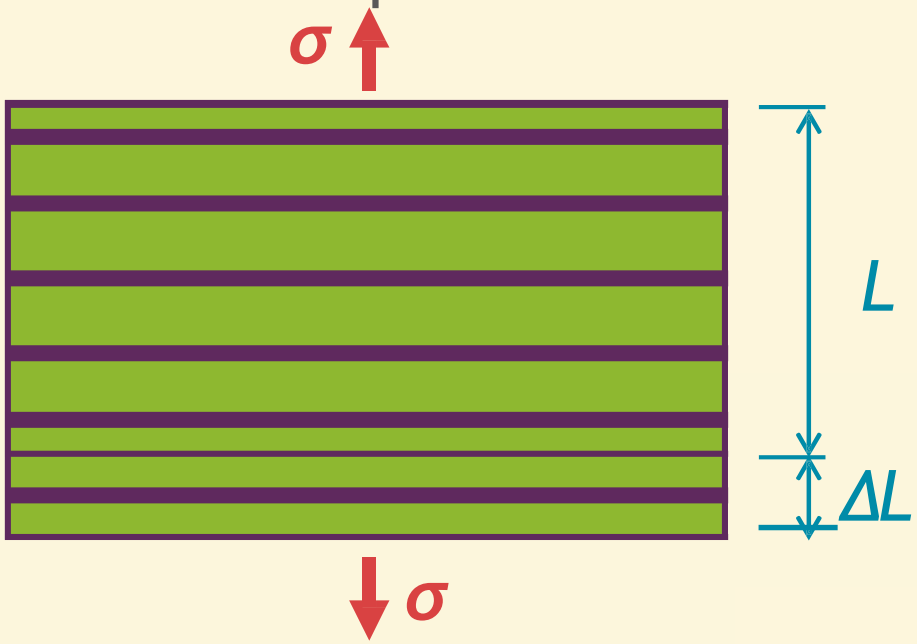
- In equal proportions, $\varepsilon_n = \left(\varepsilon_{n,f} + \varepsilon_{n,m} \right)/2$
- In un-equal proportions, $\varepsilon_n = V_f \varepsilon_{n,f} + \left(1 - V_f\right) \varepsilon_{n,m}$
- Since $\sigma = E \varepsilon_n$:
  - $\varepsilon_n = \left[\frac{V_f}{E_f} + \frac{\left(1 - V_f\right)}{E_m} \right] \sigma$

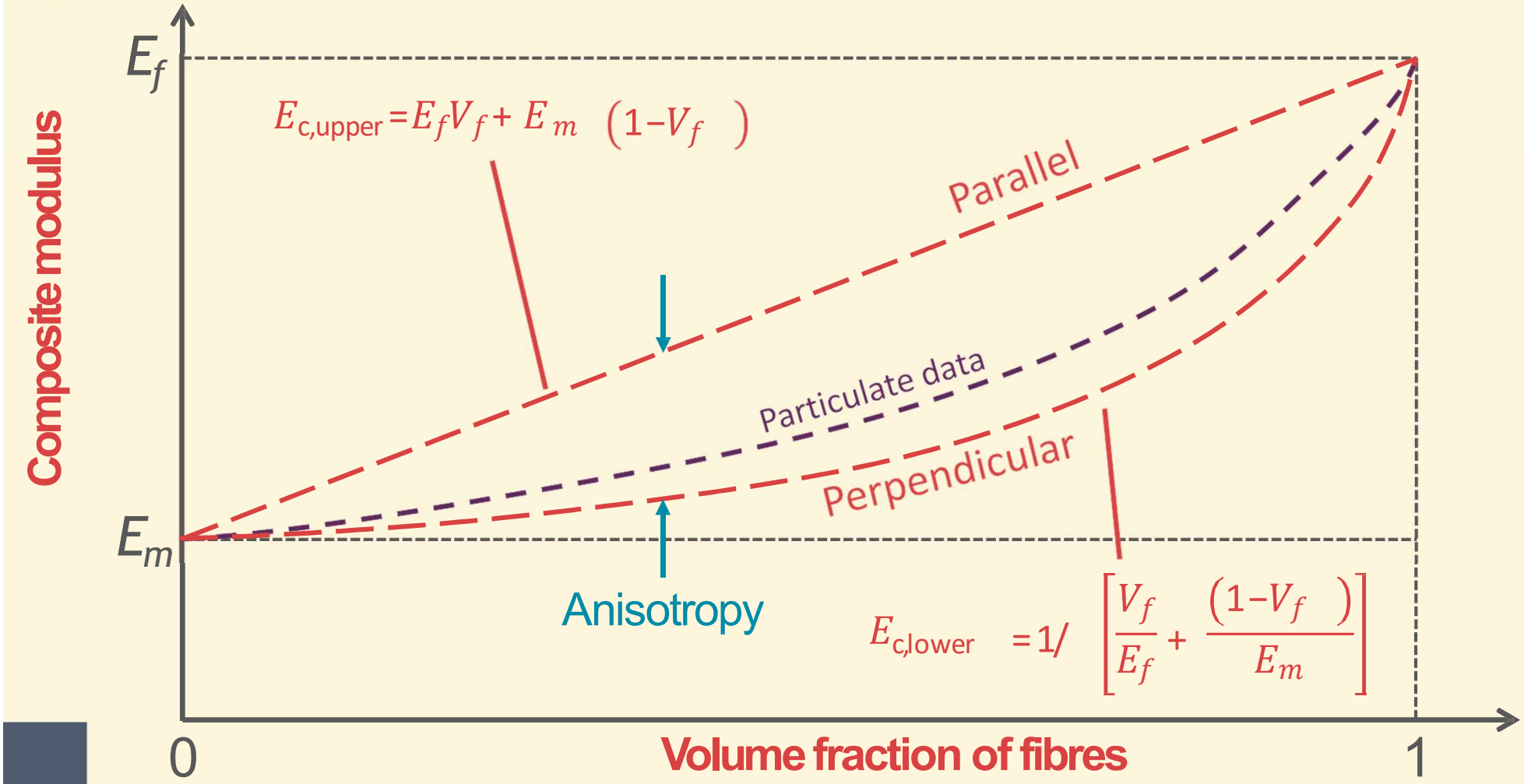


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Rule of mixtures (6)

- $\varepsilon_n = \left[\frac{V_f}{E_f} + \frac{(1-V_f)}{E_m}\right]\sigma$
- As $E_{\text{composite}} = \frac{\sigma}{\varepsilon_n}$
  - $E_{\text{composite}} = 1/\left[\frac{V_f}{E_f} + \frac{(1-V_f)}{E_m}\right]$
- **Lower estimate** for the composite modulus





Rule of mixtures (7)

- Fibre-reinforced composites are **anisotropic**:
  - Different strength depending on loading direction
- Particulate composites are **isotropic**

Rule of mixtures (8)

- **Alignment** 'efficiency':
  - Fibre orientation with respect to loading axis
  - **Warning:** Transverse properties can be *worse* than un-reinforced matrix!
  - We include **orientation factor**, η_O
    e.g. $E_c = \eta_O E_f V_f + E_m (1 - V_f)$
  - (2D random, $\eta_O = 3/8$; 3D random, $\eta_O = 1/5$)

- Overcoming anisotropy:
  - **Laminating** so alternate layers have different fibre directions